ICC - 15062026 - QUESTIONS

Status	Finished
Started	Monday, 15 June 2026, 2:06 PM
Completed	Monday, 15 June 2026, 2:20 PM
Duration	13 mins 42 secs

Question 1
Partially correct
Flag question

The following sets can be encoded as binary strings:

Select one or more:

- The set of complex numbers, i.e. $\mathbb{C}$.
- The set of triples of real numbers, i.e., $\mathbb{R}\text{\time}\mathbb{R} \times \mathbb{R}$.
- The set of pairs of rational numbers, i.e., $\mathbb{Q} \times \mathbb{Q}$.

Question 2
Correct
Flag question

The problem of checking whether two Turing machines are such that the former has less states than the latter is:

Select one or more:

- In the class $\mathbf{NP}$
- Decidable in exponential time
- Undecidable.
- Not in $\mathbf{NP}$

Question 3
Incorrect
Flag question

Which ones of the following inclusions between complexity classes is compatible with our current knowledge (i.e. maybe we do not know if this is true, but it could be)? Here, $\mathbf{NPC}$ stands for the class of all $\mathbf{NP}$-complete problems.

Select one or more:

- $\mathbf{NP} \subseteq \mathbf{NPC}$.
- $\mathbf{P} \supseteq \mathbf{NPC}$.
- $\mathbf{PSPACE} \subseteq \mathbf{L}$.
- $\mathbf{P} = \mathbf{EXP}$.

Question 4
Correct
Flag question

A friend of yours claims he designed an algorithm solving the $\mathtt{SAT}$ and working in polynomial space. You can safely conclude that:

Select one or more:

- Your friend must be right, because all algorithms solving $\mathtt{SAT}$ works in polynomial space.
- Your friend's algorithm might be correct, but one has of course to check the details.
- Even if your friend is correct, $\mathtt{SAT}$ would stay $\mathbf{NP}$-complete.
- If your friend is right, then $\mathbf{P} = \mathbf{NP}$.

Question 5
Partially correct
Flag question

The representation class of closed intervals in the real line $\mathbb{R}$:

Select one or more:

- Has infinite VC-dimension.
- Has a VC-dimension equal to that of axis-aligned rectangles in $\mathbb{R}^2$.
- Is efficiently PAC-learnable.
- Has VC-dimension equal to 2

Finish review